Que :- To study printer deadlock scenarios, analyze resource contention, and propose synchronization strategies ensuring smooth, efficient printing operations.

Code:

```html
You, 9 minutes ago | 1 author (You)
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Deadlock in Operating Systems</title>
    <link href="https://fonts.googleapis.com/css2?family=Outfit:wght@300;400;600;800&display=swap" rel="stylesheet">
    <style>
        :root {
            --primary: #6366f1;
            --secondary: #a855f7;
            --dark: #0f172a;
            --card-bg: rgba(30, 41, 59, 0.7);
            --text: #f8fafc;
        }    You, 9 minutes ago • Practical 9 OS

        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
            font-family: 'Outfit', sans-serif;
        }

        body {
            background: linear-gradient(135deg, #0f172a 0%, #1e1b4b 100%);
            color: var(--text);
            min-height: 100vh;
            overflow-x: hidden;
        }

        /* Ambient Glow Background */
        .ambient-glow {
            position: fixed;
            top: 50%;
            left: 50%;
            width: 80vw;
```

```html
        .ambient-glow {
            width: 80vw;
            height: 80vh;
            transform: translate(-50%, -50%);
            background: radial-gradient(circle, rgba(99, 102, 241, 0.15) 0%, rgba(0,0,0,0) 70%);
            z-index: -1;
            pointer-events: none;
        }

        /* Navigation Bar */
        nav {
            background: rgba(15, 23, 42, 0.8);
            padding: 1.2rem 0;
            position: fixed;
            width: 100%;
            top: 0;
            z-index: 1000;
            backdrop-filter: blur(12px);
            border-bottom: 1px solid rgba(255, 255, 255, 0.1);
        }

        nav ul {
            list-style: none;
            display: flex;
            justify-content: center;
            align-items: center;
            gap: 2rem;
        }

        nav ul li a {
            color: rgba(255, 255, 255, 0.7);
            text-decoration: none;
            font-weight: 600;
            font-size: 1.1rem;
```

```html
        .car-elem {
            font-size: 1.5rem;
            z-index: 20;
            transition: left 0.5s linear, top 0.5s linear;
            box-shadow: 0 10px 20px rgba(0,0,0,0.5);
            opacity: 0;
        }

        .car-v { width: 40px; height: 70px; }
        .car-h { width: 70px; height: 40px; }

        .car-elem.visible {
            opacity: 1;
        }

        /* 4 Conditions Cards */
        .conditions {
            display: grid;
            grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
            gap: 1.5rem;
        }

        .condition-card {
            background: rgba(15, 23, 42, 0.6);
            border: 1px solid rgba(255,255,255,0.05);
            padding: 2rem;
            border-radius: 20px;
            transition: all 0.3s ease;
        }

        .condition-card:hover {
            transform: translateY(-5px);
            background: rgba(30, 41, 59, 0.8);
            border-color: rgba(99, 102, 241, 0.3);
```

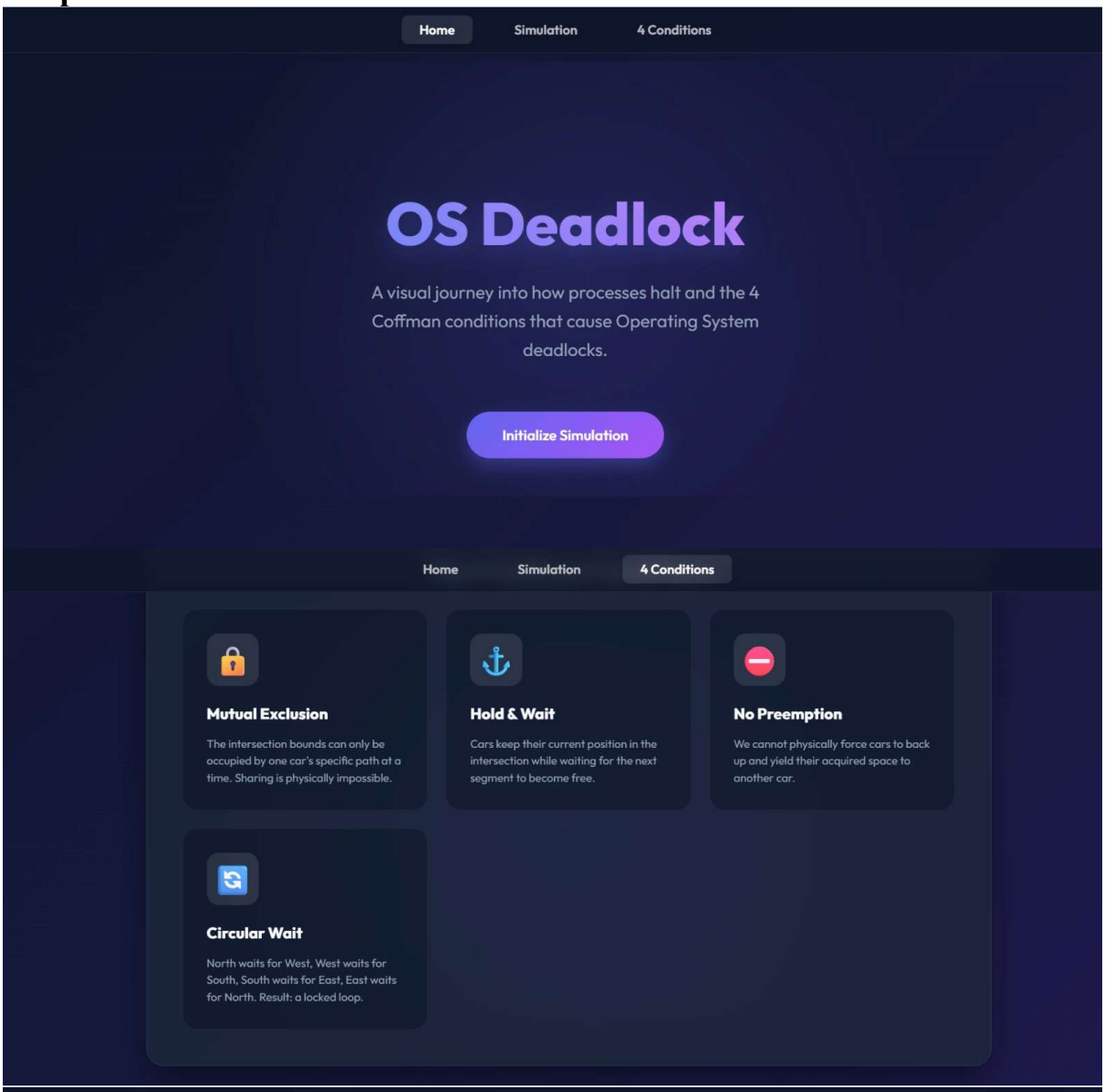
```html
<head>
    <style>
    </style>
</head>
<body>
    <div class="ambient-glow"></div>
    <div class="deadlock-overlay" id="deadlockOverlay"></div>

    <div class="alert-box" id="alertBox">
        <h2>🚨 DEADLOCK DETECTED! 🚨</h2>
        <p style="color: #cbd5e1; font-size: 1.1rem; margin-bottom: 2rem; max-width: 400px; line-height: 1.5; margin-left:auto; margin-right:auto;">
            All 4 conditions are met! Cars wait in a circular chain. No one can proceed because the intersection is permanently blocked.
        </p>
        <button class="reset-btn" style="margin: 0 auto;" onclick="resetSimulation();">
            🔄 Resolve & Reset
        </button>
    </div>

    <!-- Navigation Bar -->
    <nav>
        <ul>
            <li><a class="nav-link active" onclick="showPage('start', this)">Home</a></li>
            <li><a class="nav-link" onclick="showPage('deadlock', this)">Simulation</a></li>
            <li><a class="nav-link" onclick="showPage('deadlock', this)">4 Conditions</a></li>
        </ul>
    </nav>

    <div class="container">
        <!-- Page 1: Start Page -->
        <div id="start" class="start-page">
            <h1 class="start-title">OS Deadlock</h1>
            <p class="subtitle">
                A visual journey into how processes halt and the 4 Coffman conditions that cause Operating System deadlocks.
            </p>
            <button class="primary-btn" onclick="showPage('deadlock', document.querySelectorAll('.nav-link')[1])">
                Initialize Simulation
```

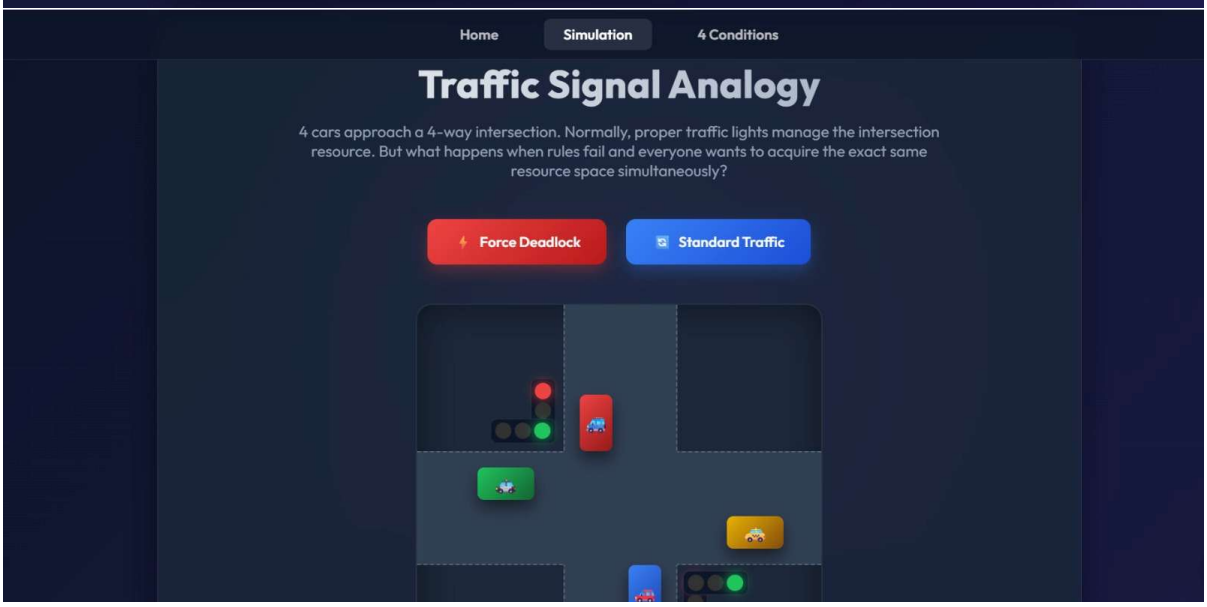
Launchpad 0 ⚠ 2 -- NORMAL -- You, 7 minutes ago Hariom Dwivedi (10 minutes ago) Ln 674, Col 8 Spaces: 4 UTF-8 LF {} HTML

```
        // Base positions for each direction [top, left]
        const CAR_POS = {
            north: { start: [20, 200], wait: [110, 200], end: [440, 200] },
            south: { start: [410, 260], wait: [320, 260], end: [20, 260] },
            west: { start: [260, 20], wait: [260, 110], end: [260, 440] },
            east: { start: [200, 410], wait: [200, 320], end: [200, 20] }
        };

        function showPage(pageId, navElement) {
            document.querySelectorAll('.start-page, .deadlock-page').forEach(page => {
                page.style.display = 'none';
            });
            document.getElementById(pageId).style.display = 'block';

            document.querySelectorAll('.nav-link').forEach(link => link.classList.remove('active'));
            if(navElement) navElement.classList.add('active');

            if (pageId === 'deadlock') {
                if(!intervalId) resetSimulation();
            } else {
                clearInterval(intervalId);
                intervalId = null;
            }
        }

        // Set light explicitly
        function setLight(dir, color) {
            ['r', 'y', 'g'].forEach(c => {
                const el = document.getElementById(`tl-${dir}-${c}`);
                el.style.background = '#333';
                el.style.boxShadow = 'none';
            });

            const target = document.getElementById(`tl-${dir}-${color[0]}`);
```

```
        function triggerDeadlock() {
            clearInterval(intervalId);

            document.getElementById('btnDeadlock').disabled = true;
            document.getElementById('btnDeadlock').style.opacity = '0.5';

            // Set all lights to red
            ['n', 's', 'w', 'e'].forEach(dir => setLight(dir, 'red'));

            // All cars move to wait point
            ['north', 'south', 'west', 'east'].forEach(dir => {
                const el = document.getElementById(`car-${dir}`);
                el.style.transition = 'left 1s cubic-bezier(0.2, 0.8, 0.2, 1), top 1s cubic-bezier(0.2, 0.8, 0.2, 1)';
                // Position so they are blocking each other directly in intersection:
                let x = CAR_POS[dir].wait[1]; let y = CAR_POS[dir].wait[0];
                if(dir==='north') y=200;
                if(dir==='south') y=260;
                if(dir==='west') x=200;
                if(dir==='east') x=260;

                el.style.top = y + 'px';
                el.style.left = x + 'px';
            });

            setTimeout(() => {
                document.getElementById('deadlockOverlay').style.display = 'block';
                document.getElementById('alertBox').classList.add('show');
            }, 1200);
        }
    </script>
</body>
</html>      You, 10 minutes ago • Practical 9 OS
```

Output:


Home
Simulation
4 Conditions
OS Deadlock
A visual journey into how processes halt and the 4 Coffman conditions that cause Operating System deadlocks.
Initialize Simulation
Home
Simulation
4 Conditions
Mutual Exclusion
The intersection bounds can only be occupied by one car's specific path at a time. Sharing is physically impossible.
Hold & Wait
Cars keep their current position in the intersection while waiting for the next segment to become free.
No Preemption
We cannot physically force cars to back up and yield their acquired space to another car.
Circular Wait
North waits for West, West waits for South, South waits for East, East waits for North. Result: a locked loop.


Home
Simulation
4 Conditions
Traffic Signal Analogy
4 cars approach a 4-way intersection. Normally, proper traffic lights manage the intersection resource. But what happens when rules fail and everyone wants to acquire the exact same resource space simultaneously?
Force Deadlock
Standard Traffic